SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – SHORT ANSWER TEST

Course Code:	CSA16	**Course Title:**	Data Warehousing and Data Mining
CO Assessed:	CO1 – Precision (BL1, BL2)	**Assessment:**	Assessment Tool 1 – Short Answer Test
Weightage:	10% of CIA	**Total Marks:**	100 (5 Questions × 20 Marks)
CO1 Statement:	*Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2*		
Student Name:	P. Giriprasad Reddy	**Reg. No.:**	199525172
Section / Batch:	AIML	**Date:**	

Instructions to Students

- Answer all questions. Each question carries 20 Marks. Total: 100 Marks.
- Clearly show all requirements, process, operations.
- Use appropriate models, schema represented scenario wise
- Justify each step in different dimension specified and measures clearly.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Short Answer Test	Architecture of Data Warehouse, ETL, OLAP	Q1 – Q5	Q1 –20 Q2 –20 Q3 –20 Q4 –20 Q5 –20
		GRAND TOTAL:	**100 Marks**

SECTION A – SHORT ANSWER TEST

1. Suppose that a data warehouse for City Hospital consists of the following four dimensions: patient, doctor, department, and time, and two measures visit_count and avg_charge. At the lowest conceptual level, avg_charge stores the actual consultation fee charged for a patient visit. At higher conceptual levels, it stores the average fee.

(a) Draw a snowflake schema diagram for the data warehouse.

(b) Starting with the base cuboid [patient, doctor, department, time], what specific OLAP operations should be performed to list the average consultation fee department-wise for each year?

(c) If each dimension has four levels (including all), how many cuboids will this cube contain (including base and apex cuboids)?

2. A smart city project is collecting data from multiple sources, including:

- Traffic sensors for vehicle counts and congestion levels.
- IoT devices for monitoring air quality, noise levels, and energy consumption.
- Public transportation systems for ridership data and punctuality.
- Citizen feedback from a mobile app.

Questions:
Fact and Dimension Design:

- Design a star schema to analyze traffic congestion patterns, energy consumption trends, and public transportation efficiency.
- What measures and dimensions would you include? Justify your design.

3. Suppose that a banking data warehouse contains two fact tables: Transaction_Fact and Loan_Fact, sharing dimensions customer, branch, time, and account_type.

- Transaction_Fact → transaction_count, amount
- Loan_Fact → loan_amount, interest

(a) Draw a galaxy schema diagram.
(b) Identify OLAP operations to obtain branch-wise average transaction amount and loan amount yearly.
(c) If each dimension has four levels, calculate total cuboids.

4. A manufacturing company monitors product quality and production efficiency. The data warehouse contains:

- Dimensions:
  - Product: Category, type, and batch.
  - Factory: Location, department, and line number.
  - Time: Year, quarter, month, and week.
  - Supplier: Region, rating, and contract type.
- Measures:
  - Defect rate.
  - Production units.
  - Supplier reliability percentage.

Questions:

- Roll-up: Summarize defect rates by product category for the current year across all factories.
- Drill-down: Focus on a specific factory to analyze defect rates by department and product type.
- Slice: Isolate production unit data for suppliers with a rating of 4+ across all regions.
- Dice: Analyze defect rates and supplier reliability for selected product categories and factory locations.

5. A retail chain manages inventory and sales data across multiple stores. The data warehouse contains:

- Dimensions:
  - Product: Category, subcategory, and SKU.
  - Store: Region, city, and store ID.
  - Time: Year, quarter, month, and day.
  - Customer: Membership tier, age group, and gender.
- Measures:
  - Inventory turnover.
  - Total sales revenue.
  - Units sold.

Questions:

1. Roll-up: Summarize inventory turnover by product category across all stores for the last quarter.
2. Drill-down: Focus on sales data for electronics in a specific region, analysing by subcategory and SKU.
3. Slice: Isolate sales revenue data for male customers in the Bronze membership tier across all stores.
4. Dice: Analyze sales revenue and units sold for selected cities and product categories during specific time periods.

Code of Conduct

I P.Giriprasad Reddy/192525172 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: [signature]

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Marks Evaluation:

Question No.	Understanding of Concepts (25%)	Explanation (25%)	Application (20%)	Organization (15%)	Accuracy (10%)	Clarity (5%)	Total Marks
Q1 (20 Marks)							
Q2 (20 Marks)							
Q3 (20 Marks)							
Q4 (20 Marks)							
Q5 (20 Marks							
					Overall Total (100)		

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ________________	Date: ________________	Date: ________________

ASSESSMENT TOOL - 01

Co1 AT1

CSA1613 - Data warehouse & Data mining

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1. Hospital Data warehouse:-

a Snowflake scheme:-

fact table :- visit - fact

Dimension:-

Patient - ID

Doctor - ID

Department - ID

Time - ID

measures:-

- visit - count

Avg - charge

Dimension Tables:

* patient_Dim → patient_ID, Name, Age, Gender
* Doctor - Dim → Doctor - D, Name, specialization
* Department_Dim → Department_ID, Dept_Name
* Time_Dim → Day, month, Quater, year

snowflake structure:

Patient_Dim
↓
Doctor - Dim —— visit_fact —— Time - DIM
|
Department - Dim

(b) OLAP operations:-

To find avg consultation fee department_wise for each year:

* Rollup :- Aggregate data from month → year
* Dice :- select specific departments and years

(c) Total Cuboids :-

Dimensions = 4

cuboids $= 2^4 = 16$ cuboids

(2) smart city Data ware house :-

measures:-

* vehicle - count
* Air - Quality index
* Congestion - level
* punctuality

Dimensions:-

1. Time Dimension:-

   * Day, month, year

2. Location Dimension:-

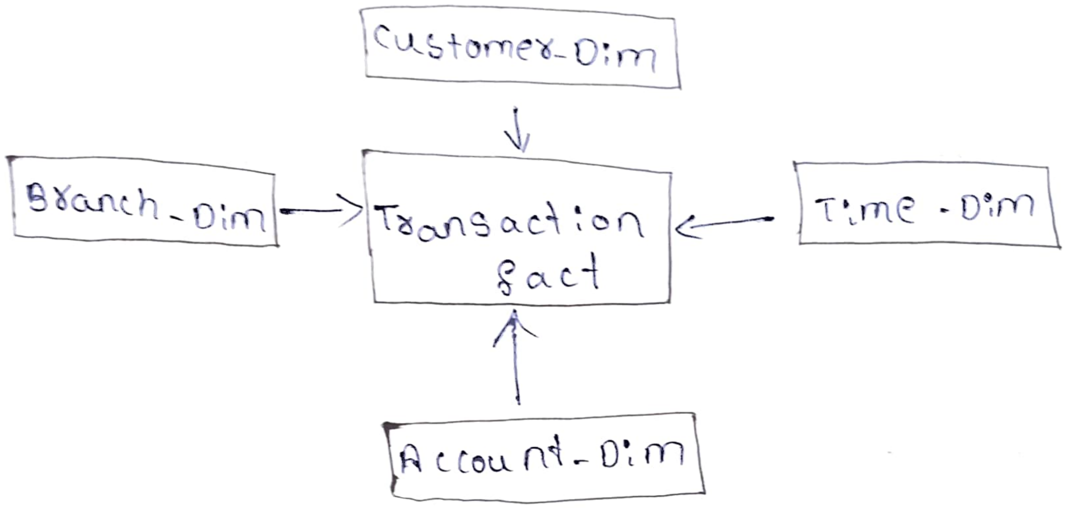
   * city, Area, zone

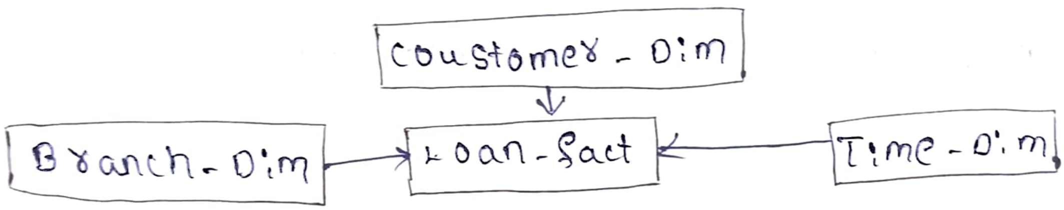
* Transport Dimension
  * Vehicle type Route

Justification:- This design helps to analyze traffic, pollution, energy usage & Transport efficiency.

3. Banking Data warehouse:-

ⓐ Galaxy Schema:-





ⓑ OLAP operations:-

For branch-wise Avg transaction & loan amount

* Roll up:- Branch - Region
* Slice :- select a particular year
* Dice :- select branches & loan types
* Pivot :- compare branches with measures.

c Total cuboids

Dimensions = 4

Cuboids $= 2^4 = 16$ cuboids

(4) Manufacturing Data warehouse:-

Rollup:-

* Defect rates by product category for current year.

operations:-

* product → category
* Time → year

Result :- Avg defect rate for each product category across factories.

Dill down:-

factory → Department → product type

Example:- factory A → production Department → electronics products.

Slice:-

select :

* supplier rating : 4+
* All regions

Dice:-

Analyze:

* selected product categories
* selected factory locations.

5. Retail chain Data warehouse:-

Roll up:-

Inventory turnover by product category

Hierarchy :- product → category

Result :- Total inventory turnover for each category across stores.

Drill down:-

Analyze electronic sales:-

Region → store → sub category → SKU

Slice :-

conditions :-

* Gender = male
* membership = Bronze

Dice:-

select:-

* specific cities
* Time periods.